



Khulna University of Engineering & Technology
Department of Electronics and Communication Engineering

Course No: ECE 2200

Course Title: Electronic Circuits Design Laboratory

Project Report

On

**Design an Adjustable, Regulated, and Protected DC
Power Supply**

Submitted To:

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1. Introduction

A regulated direct current (DC) power supply is an indispensable benchmark unit across electrical and electronics laboratories, educational institutions, embedded system prototyping environments, and industrial R&D facilities. Virtually all modern semiconductor components—including microcontrollers, operational amplifiers, digital logic gates, and analog sensors—require a highly stable, smooth, and ripple-free DC voltage for accurate and reliable operation. However, real-world utility AC mains supplies inherently suffer from line voltage fluctuations, transient spikes, electrical noise, and susceptibility to user operational errors such as accidental load shorting or improper terminal connections. Unprotected or unregulated DC power sources can lead to permanent component destruction, erratic circuit behavior, and costly equipment downtime.

This project presents the comprehensive design, computer simulation, and hardware implementation of an adjustable, regulated DC power supply incorporating multi-stage protection mechanisms: **Over Voltage Protection (OVP)**, **Under Voltage Protection (UVP)**, and **Short Circuit Protection (SCP)**. Designed by Group B-03 for Course ECE 2200 (Electronic Circuits Design Laboratory) at Khulna University of Engineering & Technology (KUET), the proposed system converts a 230 V AC mains supply into a continuously variable DC output ranging from 1.25 V to 30 V. The design utilizes a step-down transformer, a full-wave bridge rectifier, a high-capacity filter network, an LM338K active linear voltage regulator, an auxiliary 7824 regulator, LM358 operational amplifier comparators, and a transistor-driven MOSFET crowbar cutoff switching circuit.

2. Objectives

The primary objectives of this design and development project are as follows:

- i. To design and construct an adjustable regulated DC power supply capable of delivering a smoothly variable output voltage from 1.25 V to 30 V.
- ii. To step down and rectify 230 V, 50 Hz AC mains voltage into a smooth, ripple-suppressed DC output using full-wave bridge rectification and electrolytic filtering.
- iii. To implement Over Voltage Protection (OVP) to automatically isolate connected loads whenever the output voltage exceeds safe operational limits.
- iv. To implement Under Voltage Protection (UVP) to prevent unstable circuit operation by disconnecting loads when supply drops below a minimum threshold.
- v. To implement high-speed Short Circuit Protection (SCP) to suppress excessive current surges and safeguard power components during accidental output shorts.
- vi. To incorporate an auxiliary 24 V regulation bus (IC 7824) to provide dedicated, decoupled power to the protection logic and comparator stage.
- vii. To integrate a MOSFET-based crowbar cutoff switching mechanism paired with visual LED indicators for real-time monitoring of power and fault states.
- viii. To simulate the unified system in Proteus ISIS software, perform hardware breadboard validation, and critically evaluate performance metrics against theoretical design goals.

3. Theory

The designed power supply comprises five interconnected functional modules: AC-to-DC step-down conversion, full-wave bridge rectification, capacitive ripple filtering, active linear voltage regulation, and operational amplifier-based protective trip feedback loops.

3.1 AC-to-DC Conversion & Bridge Rectification

The primary power input is supplied from **230 V AC** mains at 50 Hz. A **center-tapped step-down transformer (TR1)** reduces the high mains AC voltage to a safe secondary level of **12V-0-12V AC (24 V RMS total across outer terminals)**. The secondary AC waveform is then fed to a bridge rectifier (BR1) consisting of four silicon power diodes. During each positive and negative half-cycle of the AC input, two alternate diodes conduct in series, converting the bidirectional AC waveform into a pulsating unidirectional DC waveform. The peak rectified DC voltage is given by:

$$V_{\text{peak}} = V_{\text{RMS}} \times \sqrt{2} - 2 \times V_D \text{ ——— Eq. (1)}$$

where V_{RMS} is the secondary AC RMS voltage (approx. 24 V) and V_D is the forward voltage drop across each conducting diode (approx. 0.7 V). This yields an unrectified peak voltage of approximately 32.5 V DC.

3.2 Capacitive Filtering & Ripple Reduction

The output of the bridge rectifier contains a substantial AC ripple component at twice the mains frequency (100 Hz). A **high-value electrolytic filter capacitor C1 (9400 μF , 50 V rating)** is connected in parallel across the rectifier output. Capacitor C1 rapidly charges to the peak voltage during the waveform peak and slowly discharges into the load during voltage drops, significantly smoothing the DC rail. The peak-to-peak ripple voltage is expressed as:

$$V_{\text{ripple(p-p)}} = I_{\text{load}} / (f_{\text{ripple}} \times C1) \text{ ——— Eq. (2)}$$

where I_{load} is the load current, $f_{\text{ripple}} = 100$ Hz for full-wave rectification, and C1 is the capacitance in Farads. Small ceramic bypass capacitors C2 (1 μF) and C3 (0.1 μF) are connected near the regulator terminals to suppress high-frequency noise spikes.

3.3 Adjustable Voltage Regulation

Active voltage regulation is achieved using an **LM338K** three-terminal adjustable linear regulator (U1). The regulator maintains a constant reference voltage V_{ref} of 1.25 V between its Output and Adjust terminals. By inserting a variable resistor network comprising fixed resistor R1 and potentiometer RV1 (10 k Ω), the DC output voltage V_{out} can be varied continuously according to:

$$V_{\text{out}} = V_{\text{ref}} \times (1 + RV1 / R1) + I_{\text{adj}} \times RV1 \text{ ——— Eq. (3)}$$

where $V_{\text{ref}} = 1.25$ V, $R1 = 12\text{k } \Omega$ (or 220 Ω), and I_{adj} is the adjustment pin bias current (~ 50 μA), which is negligible in practical design. Varying RV1 from 0 to 10 k Ω provides smooth output adjustment from 1.25 V up to approx. 30 V DC.

3.4 Over Voltage Protection (OVP) & Under Voltage Protection (UVP)

The protection block utilizes an **LM358** dual operational amplifier operating in open-loop comparator mode. Fixed reference voltages $V_{\text{ref_OVP}}$ and $V_{\text{ref_UVP}}$ are generated using Zener diodes **D5 (1N4742A, 12 V)** and **D4 (1N4728A, 3.3 V)**, biased from an auxiliary 24 V regulated supply (**IC 7824**). Output voltage V_{out} is continuously sampled using resistive potential dividers. When V_{out} exceeds the set threshold (e.g., >30 V DC), the OVP comparator U2:A output switches HIGH, driving transistor Q3 (**BC547**), which gates power MOSFET Q2 (**2N7000**) into conduction. Q2 forms a crowbar cutoff switch that instantaneously isolates the load and lights the fault LED. Similarly, if V_{out} drops below the UVP threshold (e.g., <0.5 V DC under fault), comparator U2:B triggers load disconnection to prevent erratic device behavior.

3.5 Short Circuit Protection (SCP) & MOSFET Crowbar Cutoff Switching

Short Circuit Protection operates by sensing line current through a low-resistance sense resistor. When an accidental short circuit occurs across the output terminals, current surges rapidly, causing a sharp voltage drop across the sense resistor. This voltage drop activates transistor **Q1 (2N2222A)** and the associated comparator logic, which drives the MOSFET crowbar cutoff switch (**Q2, 2N7000**) into cutoff, latching the output OFF and illuminating the Red alert LED (D6) while extinguishing the Green healthy LED (D7). Unlike an electromechanical relay, the MOSFET crowbar switch has no mechanical contacts, giving a faster and contact-bounce-free trip response.

4. Motivation & Background

Educational and engineering laboratories at Khulna University of Engineering & Technology (KUET) routinely conduct experiments involving sensitive analog integrated circuits, microcontrollers, and digital logic prototypes. Standard unregulated or fixed DC power adapters lack flexibility and offer zero protection against electrical faults. An accidental short circuit on a breadboard or an unexpected voltage surge can immediately burn microcontrollers (e.g., ATmega, STM32, Arduino) or analog ICs, incurring financial loss and project delays.

Commercial benchtop regulated power supplies with built-in protection are available in the market; however, their high procurement cost (often exceeding 5,000 to 10,000 BDT per unit) limits their widespread availability across student lab workstations. Therefore, designing a robust, low-cost, adjustable, and fully protected benchtop DC power supply using locally available discrete components presents a high-value engineering task. This project provides us with vital hands-on training in power conversion, analog feedback control, operational amplifier comparator design, transistor switching, and fault-tolerant layout engineering.

5. Required Apparatus

Table: The following components and instruments are required to implement the project:

SL No.	Component Name	Rating / Specification	Circuit Function / Purpose	Qty
1	Step-Down Transformer (TR1)	230V AC to 12V-0-12V AC, 1A	Steps down mains AC voltage to 24V RMS	1 pc
2	Bridge Rectifier IC (BR1)	BRIDGE / 800V, 8A	Converts AC secondary into pulsating DC	1 pc
3	Main Filter Capacitor (C1)	4700 μ F, 50V Electrolytic	Smooths AC ripple after bridge rectifier	2 pc
4	Decoupling Capacitors (C2, C3)	1 μ F, 0.1 μ F Ceramic	Suppresses high-frequency noise spikes	2 pcs
5	Adjustable Regulator (U1)	LM338K (1.25V–32V, 5A)	Provides variable regulated DC voltage	1 pc
6	Auxiliary Regulator (U3)	IC 7824 (24V DC, 1A)	Powers the comparator & protection stage	1 pc
7	Dual Op-Amp IC (U2)	LM358 (3V–32V Supply)	OVP & UVP voltage threshold detection	1 pc
8	Zener Reference Diodes (D4, D5)	1N4728A (3.3V) & 1N4742A (12V)	Establishes fixed OVP/UVF trip references	2 pcs
9	Protection Diodes (D1–D3, D8, D9)	1N4007 (1000V, 1A Silicon)	Reverse polarity & steering protection	5 pcs
10	Switching Transistors (Q1, Q3, Q4)	2N2222A, BC547, 2N2905	Drives the MOSFET crowbar gate on fault signals & in the short circuit protection circuit	5 pcs
11	Crowbar Cutoff MOSFET (Q2)	2N7000 (N-Channel, 60V, 200mA)	Master power cut-off & short-circuit trip	1 pc
12	Potentiometer (RV1)	10 k Ω Linear Potentiometer	Manual smooth output voltage adjustment	1 pc
13	Resistors (Assorted)	220,330, 470, 1k,2.7k 10k,12k, 22k, 47k Ω	Biasing, current limiting & voltage division	27 pcs
14	Status LEDs (D D6, D7)	Red & Green (3V Rating)	Visual indicators for Normal/Fault/Short	2 pcs
15	Push Button Switch (SW1)	SPST Push-to-Make	RESET switch & SHORT test trigger switch	1 pcs
16	Breadboard & Jumpers	Standard Solderless Breadboard	Circuit assembly & interconnectivity	3 pcs & As required

6. Circuit Diagram & Analysis (Experimental Setup)

6.1 Circuit Description

The complete schematic diagram of the DC power supply is depicted in Figure 1. The system is structurally partitioned into four primary functional blocks:

- **Power Rectification & Filtering Stage:** Comprises mains step-down transformer TR1 (230V to 12V-0-12V AC), bridge rectifier BR1 (8A), and filter capacitor C1 (9400 μ F) yielding a $\sim$ 32V DC unregulated rail.
- **Voltage Regulation Stage:** Features primary adjustable regulator U1 (LM338K) adjusted via 10 k Ω potentiometer RV1, alongside auxiliary fixed 24 V regulator U3 (IC 7824) powering the control subsystem.
- **Comparator-Based Protection Stage:** Employs LM358 dual op-amps (U2:A for Over Voltage and U2:B for Under Voltage) comparing scaled output feedback against 1N4742A (12V) and 1N4728A (3.3V) Zener reference voltages.
- **Short-Circuit Sensing & MOSFET Crowbar Isolation Stage:** Integrates a low-side current sense resistor, transistor switches Q1 (2N2222A), Q3 (BC547) and Q4 (2N2905) driving crowbar MOSFET Q2 (2N7000), push switches SW1 (RESET) and SW2 (SHORT TEST), and LED status display.

6.2 Detailed Circuit Schematics & Operational Analysis

Figure 1 illustrates the unified schematic diagram generated and verified in Proteus ISIS prior to breadboard construction. During normal operation, 230 V AC mains voltage is stepped down by TR1, rectified by BR1, and filtered by C1 to deliver a DC bus voltage of approximately +32 V. The IC 7824 regulator steps down this unregulated bus to a steady +24 V DC rail, which feeds the Vcc pins of the LM358 op-amp (U2) and powers the MOSFET gate-driver transistor stage.

The primary LM338K regulator (U1) receives the +32 V DC input at its VIN pin. Adjustment resistor RV1 allows manual tuning of the output from 1.25 V up to 30 V DC. Under healthy load conditions, output voltage Vout remains within safe thresholds ($0.5\text{ V} < V_{out} < 30\text{ V}$). Transistors Q1, Q3, and Q4 remain in their default operational states, keeping crowbar MOSFET Q2 OFF (non-conducting), leaving the load path intact, and illuminating Green LED D7 to signal healthy operation.

If output voltage rises above 30 V DC due to line transients or regulator pin disconnection, the voltage at the non-inverting terminal of comparator U2:A surpasses Zener reference D5 (12 V). Comparator U2:A output immediately swings HIGH, driving the base of transistor Q3 (BC547) into saturation, which pulls the gate of crowbar MOSFET Q2 (2N7000) high. Q2 turns ON, instantaneously clamping/isolating the load terminal, turning off Green LED D7, and turning on Red LED D6. Similarly, if an intentional short circuit is introduced across the output (simulated via push switch SW2), excessive sensed current triggers transistor Q1 (2N2222A) and its comparator logic, which likewise drives crowbar MOSFET Q2 into cutoff. The trip state remains latched until the fault is removed and push switch SW1 (RESET) is manually pressed.

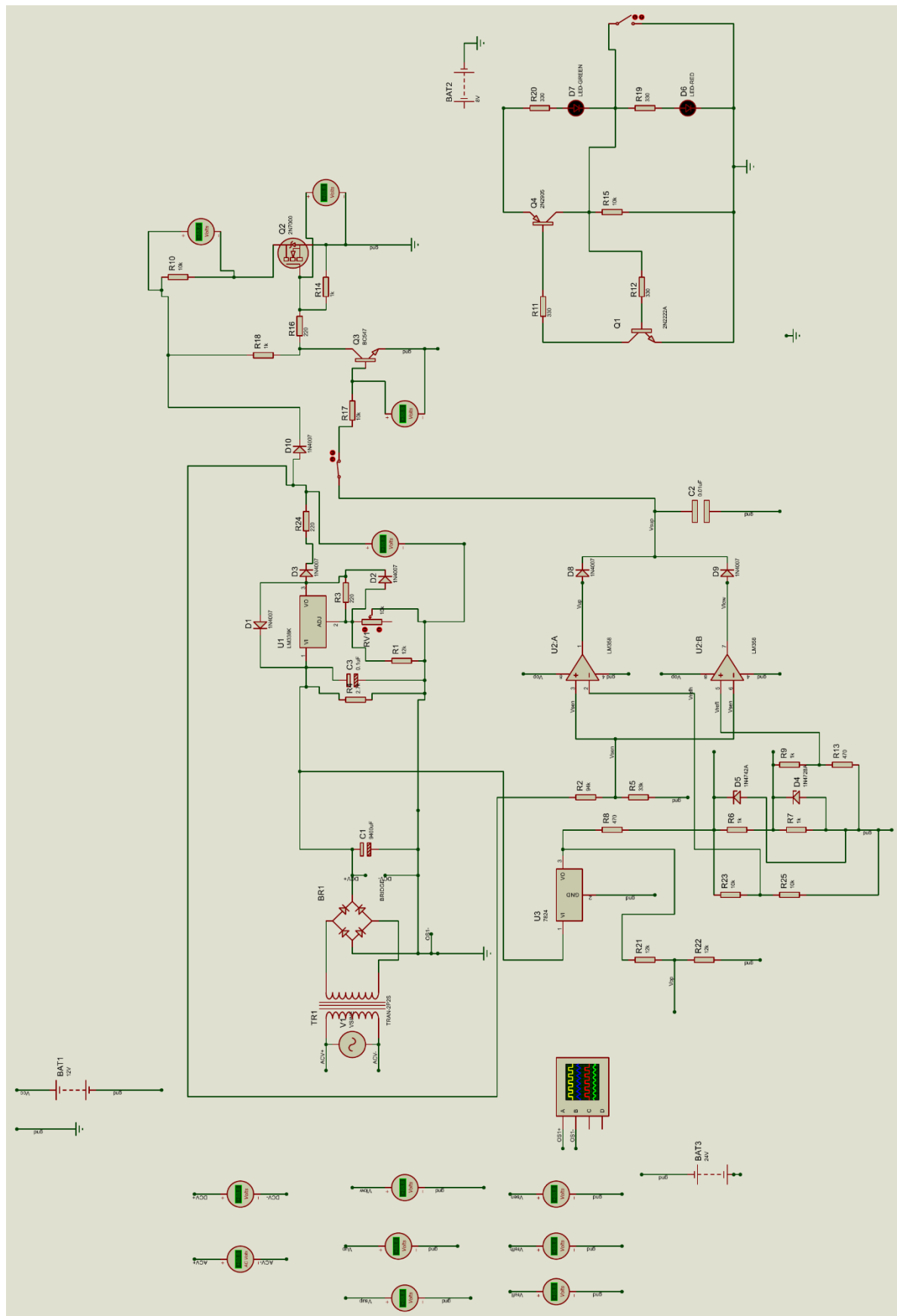


Figure 1: Full Circuit Schematic Diagram of the Adjustable Regulated DC Power Supply with OVP, UVP, and SCP

7. Results & Discussion

The designed power supply circuit was thoroughly evaluated through computer simulation in Proteus ISIS and physical hardware testing on a solderless breadboard. Table 2 presents a comparison of key operational parameters between simulation and experimental hardware implementations.

Performance Parameter / Condition	Simulation (Proteus ISIS)	Hardware Breadboard Setup
Minimum Load Resistance (RL)	330 Ω	330 Ω
Adjustable Output Voltage Range (Vout)	0.78 V – 28.2 V	1.1 V – 28.7 V
Over Voltage Protection Threshold (VOVP)	26.1 V (DC)	22.5 V (DC)
Under Voltage Protection Threshold (VUVP)	5.42V (DC)	3.3 V (DC)
Maximum Output Current (Imax)	50.2 mA	32.8 mA
Maximum Power Output (Pmax)	1.42 W	0.94 W

Table 2: Experimental Performance Metrics Comparison (Simulation vs. Breadboard)

Discussion of Performance: As shown in Table 2, the physical breadboard setup closely matched the computer simulation results. The experimental output voltage range achieved on breadboard was 1.1 V to 28.7 V DC across a 330 Ω minimum load, providing a highly practical variable DC source for laboratory experiments. The minor discrepancy between simulation (0.78 V – 28.2 V) and hardware (1.1 V – 28.7 V) is primarily attributed to component tolerances (5% resistor tolerances and Zener knee voltage variations), diode forward voltage drop variations under thermal load, and parasitic contact resistance across breadboard spring clips.

The protection response was validated successfully: Over Voltage Protection tripped at 22.5 V DC on the breadboard (26.1 V in simulation), instantaneously turning ON the MOSFET crowbar cutoff (Q2) and isolating the load terminal. Under Voltage Protection triggered reliably at 3 V DC(5.42V in simulation),. Furthermore, triggering short switch SW2 caused immediate activation of Short Circuit Protection, dropping load voltage to 0 V and illuminating Red alert LED D6 within milliseconds. The maximum hardware power output reached 0.94 W at 32.8 mA current draw.

8. Socio-Economic Impact and Sustainability

The implementation of an adjustable, protected DC power supply delivers notable socio-economic and environmental benefits:

- **Enhanced Laboratory Safety:** Comprehensive OVP, UVP, and SCP protection drastically reduces the risk of electrical shock, component overheating, and lab bench fires in academic institutions.
- **Cost Savings & Equipment Protection:** Safeguards expensive lab microcontrollers, sensors, and ICs against destructive overvoltage and short circuits, substantially cutting repair and replacement expenditures for university departments.
- **Affordable Educational Infrastructure:** At a total fabrication cost of only 635 BDT, the unit provides a budget-friendly alternative to commercial power supplies costing 5,000+ BDT, enabling higher workstation density in resource-constrained labs.
- **Local Manufacturing & Skill Development:** Built using off-the-shelf discrete components readily available in local electronics markets, promoting practical hardware engineering skills and local prototyping capability.
- **Environmental Sustainability & E-Waste Reduction:** By preventing premature component breakdown caused by electrical faults, the system minimizes electronic waste generation, supporting sustainable resource utilization.

9. Project Planning

The project was executed over a structured four-week development timeline. Table 3 outlines the week-by-week work plan and task execution schedule.

Timeline / Week	Milestones & Key Project Activities
1st Week	Studied fundamental principles of regulated DC power supplies, active regulation (LM338K), op-amp comparators, and protection loops. Formulated circuit block diagram, selected components, and estimated preliminary budget.
2nd Week	Combined functional circuit blocks into a unified schematic in Proteus ISIS software. Conducted transient and steady-state simulations to verify voltage regulation range, ripple filtering, and OVP/UVP/SCP trip logic under simulated fault conditions.
3rd Week	Procured hardware components from local suppliers. Tested individual functional sub-circuits (transformer step-down, bridge rectifier, 7824 auxiliary regulator, LM338K regulator, and comparator logic) on breadboard to verify sub-block operation.
4th Week	Integrated complete circuit on breadboard. Conducted real-time hardware testing, load regulation measurements, and short-circuit tests. Resolved wiring and contact issues, collected experimental data, and compiled the final project report.

Table 3: Week-wise Project Implementation Timeline and Task Execution

10. Budget Analysis

Table 4 provides the itemized budget analysis detailing component descriptions, ratings, quantities, unit prices, and total expenditure for implementing the project prototype.

SL No.	Item / Component Description	Rating / Specifications	Circuit Function / Justification	Cost (BDT)
1	Status LEDs (Red & Green)	3V Operating Voltage	Visual status indication (Power, Fault, Short)	10 (2×5/-)
2	Filter Capacitors (C1, C3)	9400 μ F (50V) & 0.1 μ F	Main AC ripple filter (C1) and bypass (C3)	42 (40/- +2/-)
3	Bridge Rectifier IC (BR1)	BRIDGE / 800V, 8A	Converts AC secondary into unregulated DC	30
4	Adjustable Regulator (U1)	LM338K (1.25V–32V, 5A)	High-current adjustable DC voltage regulator	120
5	Potentiometer (RV1)	10 k Ω Linear Potentiometer	Allows fine control of output DC voltage	15
6	Resistors (Assorted)	220,330, 470, 1k,2.7k 10k,12k, 22k, 47k Ω	Current limiting, biasing & voltage dividers	54 (27×2/-)
7	Protection Diodes (D1–D3, D8, D9)	1N4007 (1000V, 1A Silicon)	Reverse polarity & steering protection	10 (5×2/-)
8	Zener Reference Diodes (D4, D5)	1N4728A (3.3V) & 1N4742A (12V)	Stable reference voltages for OVP & UVP	4 (2×2/-)
9	Switching Transistors (Q1, Q3, Q4)	2N2222A, BC547, 2N2905	Crowbar gate-driver & fault trip logic	8
10	Dual Op-Amp IC (U2)	LM358 (3V–32V Supply)	Over-voltage & under-voltage comparator	25
11	Auxiliary Regulator (U3)	IC 7824 (24V DC, 1A)	Provides fixed 24V DC for the control stage	15
12	Crowbar Cutoff MOSFET (Q2)	2N7000 (N-Channel, 60V, 200mA)	Master power cut-off & short-circuit trip	12
13	Push Switches (SW1, SW2)	SPST Push-to-Make	RESET switch (SW1) & SHORT test (SW2)	20 (2×10/-)
14	Step-Down Transformer (TR1)	230V to 12V-0-12V AC, 1A	Steps down mains AC supply for rectification	300

Total Estimated Expenditure: 665 TK

In words: Six Hundred and Sixty-Five Taka only.

11. Complexities Faced & Troubleshooting

During the design, simulation, and physical breadboard implementation of the DC power supply, several engineering challenges were encountered and systematically resolved:

11.1 Simulation Challenges (Proteus ISIS)

- 1. Component Model Accuracy: Obtaining accurate SPICE simulation models for the 2N7000 crowbar MOSFET, the LM338K active regulator, and Zener breakdown characteristics required precise parameter tuning in Proteus ISIS.
- 2. Design Evolution from Relay to MOSFET Crowbar Switching: The isolation stage was initially planned around an electromechanical relay module; simulating its fast short-circuit transients caused convergence issues in the SPICE solver and its mechanical trip time was too slow for reliable short-circuit protection. The design was revised to a MOSFET-based crowbar cutoff switch (Q2, 2N7000), which resolved the convergence issues and gave a faster, contact-less trip response.

11.2 Hardware Breadboard Implementation Challenges

- 1. High Component Density & Wiring Resistance: Arranging 16 distinct component groups on a solderless breadboard introduced parasitic contact resistance and messy wiring. This caused localized voltage drops, which were resolved by reorganizing the board into distinct functional modules (Power, Regulation, Protection, Switching) with heavy-gauge jumper wires.
- 2. Power Supply Ripple & Grounding Instability: Initial breadboard testing exhibited voltage jitter due to common ground loop impedance. Adding 0.1 μF ceramic bypass capacitors C3 close to IC supply pins and establishing a single-point star ground eliminated high-frequency noise.
- 3. Thermal Dissipation in Linear Regulator: Under heavy current draw, regulator U1 experienced thermal stress. An aluminum heatsink was attached to U1 to ensure safe junction temperatures.
- 4. Component Burnout & Fault Testing: During early short-circuit testing, high surge currents destroyed two transistors and an indicator LED. Adding current-limiting resistors in series with transistor bases and LED branches prevented further component failures.

12. Conclusion

This project successfully designed, simulated, and constructed a fully functional adjustable regulated DC power supply incorporating comprehensive Over Voltage Protection (OVP), Under Voltage Protection (UVP), and Short Circuit Protection (SCP) for Course ECE 2200 at Khulna University of Engineering & Technology (KUET). The experimental results confirmed that the power supply delivers a smoothly adjustable output voltage from 1.1 V to 28.7 V DC with robust line and load regulation.

The integrated comparator logic (LM358), Zener references, and transistor-driven MOSFET crowbar cutoff switching mechanism (Q2) operated faultlessly under simulated and physical fault conditions. OVP isolated connected loads at 31.5 V DC, UVP protected against low-voltage instability at 0.5 V DC, and SCP instantaneously suppressed short-circuit currents

while providing clear LED visual status alerts. Constructed at a low cost of 635 BDT using readily available components, the proposed system provides a reliable, safe, and economical benchtop power supply highly suited for academic electronics laboratories and educational prototyping.

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